Preserving Cultural Heritage with 3D Scanning Technologies: An Overview of Methods, Their Applications and Comparative Evaluation

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Abstract

The preservation of cultural heritage using computer technologies is becoming a comprehensive discipline to study. 3D scanning technology, in particular, introduces non-contact preservation methods, namely photogrammetry, structured light, laser triangulation, computed tomography, and LiDAR, allowing documentation of delicate artifacts. Each 3D scanning method includes its own principles, case studies, and comparative efficiency in terms of accuracy, speed, portability, and applicability to natural materials. For virtual or physical replication of 3D scanned artifacts, the multimodal pipeline includes the optimal production of virtual museums or tangible replicas. This study provides an extensive overview of 3D scanning techniques for advanced methods of preserving cultural heritage, enhancing digital representation and promoting cultural globalization. This work is important for preserving cultural artifacts from potential loss, highlighting the value of digital technologies in cultural preservation.

Keywords: 3D scanning, cultural heritage, digital twin, photogrammetry, virtual reality, cnc milling

Cultural heritage includes various art forms from different time periods, such as paintings, architecture, statues, musical instruments, textiles, ceramics, tools, and other traditional artifacts. These tangible elements symbolize historical information, articulate cultural values, explain relations between cultural groups across time, and exhibit unique aesthetics. For example, as illustrated in **Figure 1**, the ceramic statue of an Uzbek wise man holding pomegranates visualizes the cultural tradition of welcoming guests.

Figure 1. Uzbek wise man. Image taken by the author.

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According to UNESCO (2025), "Heritage is our legacy from the past, what we live with today, and what we pass on to future generations. Our cultural and natural heritage are both irreplaceable sources of life and inspiration." (para.2) However, many cultural heritage sites and artifacts are at risk of deterioration, natural disasters, or acts of vandalism, resulting in the vanishing of our historical values, the loss of common memories, and the fragmentation of relationships. For this reason, the conservation and restoration disciplines are concerned with preserving cultural heritage at risk, including damaged vases, faded artworks, and deteriorated mosques and churches. These disciplines may involve "any methods that prove effective in keeping that property in as close to its original condition as possible as long as possible" (Walston, 1978, p.9). In 1972, the World Heritage Convention was established to connect countries around the globe to encourage the identification, protection, and preservation of cultural and natural heritage.

Between the 1950s and mid-1990s, the preservation of cultural heritage was predominantly manual. However, from 1996, conservation practices began relying on the use of digital technologies. The digitization is a progressive, technological approach used for preservation, restoration, education, and globalization of cultural heritage. In contrast to the material recreation of heritage or duplication of its form using alternative chemical materials, digitization implements advanced computer technologies to copy heritage without physical contact (Guidu & Frischer, 2020). This approach, commonly named as 3D scanning, produces reliable data that can be used to create digital twins or three-dimensional models of artifacts (Kantaros et al., 2021).

Based on this, digitization of cultural heritage brings such benefits as aiding the restoration process of damaged, fragile, or destroyed pieces or ones in inaccessible locations. Many cultural heritage sites are at great risk of disappearing in the modern world due to possible damage from external factors: natural disasters and erosion, wars with air attacks, tourism threats, or forms of vandalism (Benchekroun & Ullah, 2021). For example, the state of the Great Wall of China deteriorated over time, and thanks to digitization, its current state was analyzed to aid in the restoration efforts. Another example is the destruction of two massive Buddhas in Bamiyan, Afghanistan, installed inside a cliff and reaching 35 and 53 meters. Figure 2 illustrates a photo of the smaller Buddha. In 2001, the Taliban completely destroyed both statues, but conservation experts, participating in the global Buddha project, intend to restore them using 3D scanning (Kantaros et al., 2021).

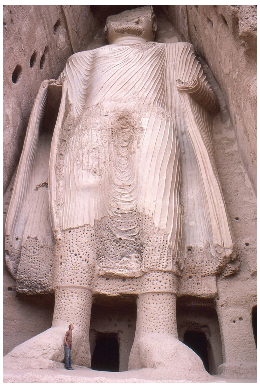

Figure 2. The smaller Buddha of Bamiyan.

Photograph by Philippe Marquis / DAFA, *BamyanBuddha Smaller 1* (1977), Wikimedia Commons, https://commons.wikimedia.org/wiki/File:BamyanBuddha_Smaller_1.jpg. In the public domain.

In addition to the restoration of damaged heritage, digitization practices also allow specialists to accurately reconstruct damaged artifacts to their original condition using 2D photographs (Qader & Shwan, 2024). For example, as Parfenov and colleagues (2022) explained, 3D scanning is not only used for virtual copying practices, but also to add broken details, remodel the damaged sections, or polish the artifact, removing micro scratches on the artifact's surface. Afterwards, a 3D printer or computer numerical control (CNC) machine can be used, depending on the original material of the artifact, to produce its physical replica. As a result, the damaged original piece can be replaced with its precisely reproduced copy for archival purposes, reducing the risks of fatal loss.

In parallel to conversation, digitization facilitates access to cultural heritage artifacts that are in inaccessible places or are physically fragile. These challenges can be addressed by 3D scanning real objects and creating their 3D copies that can be potentially displayed in virtual museums. For example, as Kantaros et al., (2021) explained, 3D scanning technology was used to save the tomb of the ancient Egyptian queen Nefertari, which was closed to the public due to preservation concerns. Similarly, the Lascaux Caves, which contain ancient and fragile artwork, were 3D scanned for conservation. 3D scanning technology helped protect the delicate artwork and later allowed the creation of virtual tours around the world. Additionally, in combination with 3D printing, the British Museum in London utilized 3D scanning technology to manufacture a replica of the fragile Rosetta Stone: while the original artifact is vulnerable and susceptible to light degradation, the 3D printed copy has no such limitations and can be tangibly observed by visitors.

Moreover, in academic and analytical fields, cultural digitization can aid in research analysis thanks to its advanced accuracy. For example, in 1901, archaeologists found a 2000-year-old device in Greece; it consisted of complex gearing systems that led scientists to conclude that this device was used to predict eclipses and other astronomical events, but still, there was no evidence or clear explanation of exactly how. However, scientists from University College London have finally solved the mystery using 3D scanning and modeling: they scanned 82 fragments of the device that survived until now, which made up about a third of the entire object, and using an ancient Greek mathematical method, modeled the device (BBC News, 2021). Other examples mentioned by Kantaros et al. (2021) were the use of 3D scanning at Notre Dame Cathedral in Paris and the Tower of London in the UK to create detailed digital twins. In other instances, the popular Sistine Chapel in the Vatican, the Parthenon temple in Athens, and the Mayan Temples of Tikal were also 3D-scanned to create detailed 3D models for further analysis and revelation of previously hidden details.

The primary objective of this research paper is to demonstrate how cultural heritage can be preserved through the implementation of 3D scanning technologies. Therefore, this study introduces 3D scanning techniques, explains how these 3D scanning methods work, and illustrates successful applications of 3D scanning in a set of international examples. In addition, this paper presents an evaluation of each 3D scanning technology, underscoring their main strengths and weaknesses, and investigates their potential applications in cultural preservation.

Literature Review

3D scanning is a complex technological process with wide-ranging applications. As Guidu & Frischer (2020) explained, the process collects necessary information about an object's geometry (shape, structure, silhouette), texture (roughness, metalness, fabric weave), color (natural, base color), scale and proportion (real and accurate size), damage or erosion (if object has, it can show cracks, chipped edges), and contextual information (importance and relationship of object to surroundings). As a result, a 3D model saves every visual characteristic of the original site. These informative resources are also called 3D annotations, and without these materials, it is impossible to replicate the artifact in the digital world. These 3D models can be used in a wide range of applications such as cultural preservation, entertainment, healthcare, manufacturing design, environmental sciences, and VR/AR.

The evolution of 3D digitization

The first experiment of 3D digitization was performed in 1996 by the Centre de Recherche et de Restauration des Musées de France (Guidu & Frischer, 2020). The Centre obtained permission to apply their innovative 3D scanning technology, specifically laser triangulation, on a painting by Jean-Baptiste-Camille Corot at the National Research Council of Canada. In 2002, scientists applied 3D scanning to paintings for the second time during the observation of "The Adoration of the Magi," a work by Leonardo da Vinci. The early 3D scanned objects were mainly traditional paintings and artworks, as the poor preservation of canvas often led to erosion of painted surfaces. During the scanning process, the typical lateral resolution was 270 µm for the entire painted surface, 90 µm for the central high-resolution area, and 350 µm for the reverse side. Additionally, Leonardo da Vinci's most well-known and iconic Renaissance painting —Mona Lisa— was 3D scanned for preservation purposes. During the process of scanning, scientists were able to study artistic techniques such as "sfumato," brushstroke patterns, and color information in an unprecedented way (Guidu & Frischer, 2020).

In 1980, traditional methods of copying sculptural monuments involved using cement mixed with marble powder, with the sculptures manually shaped by cutting. In contrast, there are now many available materials, such as wood and plastic, used to manufacture sculpture replicas, often referred to as "artificial stone." This is achieved by utilizing innovative 3D printers and CNC machines to systematize and optimize the resources that are spent in traditional copying (Parfenov et al., 2022).

The combination of 3D scanning and CNC machining technologies also represents a groundbreaking approach for the restoration practices of destroyed monuments. This task is particularly challenging because of the complexity of the shapes of the broken monuments in traditional copying and the lack of accuracy in manual reconstruction. For this reason, CNC machining (mills, grinders, and lasers) introduces automated manufacturing methods, ensuring higher quality, faster processing, and intelligent systems (Goodwin University, 2025). These machines operate based on Computer-Aided Design (CAD) models and execute machinery languages such as G-code and M-code to produce technically accurate physical replicas.

As an example, in 2002, the marble sculpture "Adam" by Tullio Lombardo was 3D scanned after a sudden incident in which the statue shattered into hundreds of pieces at the Metropolitan Museum in New York. The scanning aimed to initially collect all the pieces into a single 3D model and further enable a complete physical reconstruction (Guidu & Frischer, 2020). Another example of successful 3D scanning is the "Holy Christ of the Blood" statue. The body of the statue was nearly destroyed during the Spanish Civil War, shattering into 30 pieces. These body fragments were collected by the artist Clemente Cantos who recomposed the statue without the head. A complete manual restoration was carried out by the sculptor Juan González in October 1939. Later, in 2021, researchers Melendreras Ruiz et al. (2023) recreated a 3D model of the statue and evaluated scanning techniques for similar restoration practices.

An Overview of 3D Scanning Technologies

The overall process of 3D scanning is non-contact, which means it follows a different approach and presents advantages over traditional copying. For example, the "Primavera" statue was the first example of using non-contact copying of sculptural monuments in Russia. The scientists of St. Petersburg Electrotechnical University, in collaboration with Italian colleagues, used 3D scanning and a CNC milling machine to manufacture an artificial stone of the statue with a level of detail that is unrealistic and unattainable by manual techniques (Parfenov et al., 2022). In early 2000, the digital scanning method was ported to ancient and ruined heritage monuments. For example, a group headed by Marc Levoy started the Stanford Digital Michelangelo Project in 1998, in which 20 sculptures were scanned, including Michelangelo's most famous 5m-tall statue of David for preservation purposes (Guidu & Frischer, 2020). These case studies showed the potential use of computer systems instead of manually sculpting the statue again.

Despite the advantages of 3D scanning, there are four crucial drawbacks. First, it involves complex software and expensive equipment. Second, 3D scanning requires time-consuming computational simulations that can last for many hours. Third, the inaccuracy of scanned details should sometimes be manually cleaned or repainted on the computer. Finally, the most vital limitation depends on the material of the scanned objects: materials such as bronze and marble can reflect light in a complex way that a computer cannot calculate accordingly, so the covering of these fragments with a matte spray should be double-checked (Barszcz et al., 2023). This limitation was first noticed by the Visual Information Technology group at NRC Canada, which found that the sponge-like structure of marble can cause reflections from internal layers of the material, thereby distorting the light reflected back (Guidu & Frischer, 2020).

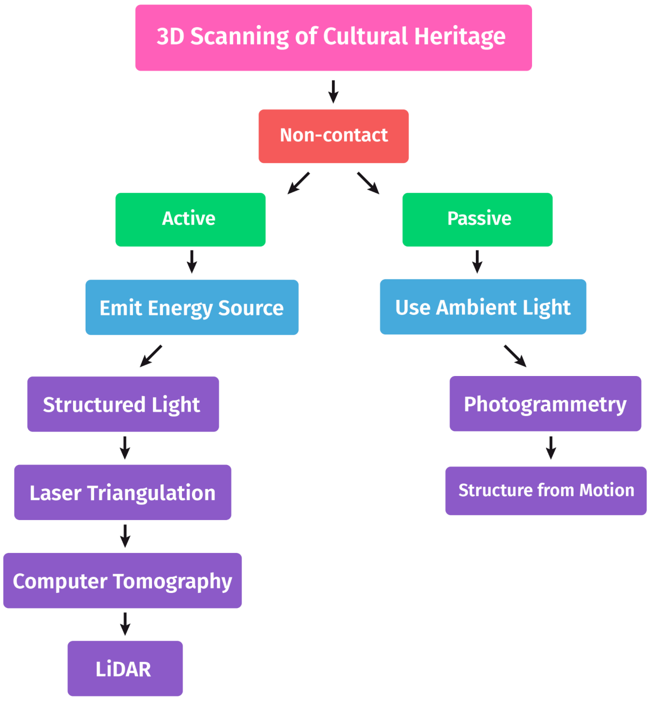

Figure 3. Hierarchical Diagram of 3D Scanning Techniques
A hierarchical diagram illustrating various 3D scanning techniques used in digital cultural heritage preservation, grouped into active and passive methods. Created by the author.

3D non-contact scanning techniques are divided into two groups: active and passive, as illustrated in **Figure 3**. The difference between these two groups is how they interact with an object. Active scanning methods may emit energy in different types, such as x-ray, light, laser, or structured pattern, to record data by measuring the distance to the object (Guidu & Frischer, 2020). These active scanning methods also include: structured light, laser triangulation, computer tomography, and LiDAR. In contrast, passive scanning methods mainly interact with the object using ambient light, multiple camera views, and mathematical trigonometry (Hermary, 2024). The main passive scanning method is photogrammetry, with its additional components like Shape-from-Motion (SfM). For individual heritage objects, each scanning technology features specific potentials and limitations depending on their scale, color, and complexity of the shape (Qader & Shwan, 2024). The next section contains detailed explanations and case studies featuring the application of each 3D scanning technique in cultural heritage preservation. At the end of the section, **Table 1** presents the optimal use, benefits, and limitations in a side-by-side comparison of each method.

Photogrammetry

Photogrammetry is the process of computer simulation that creates a 3D model based on 2D photographs taken from different angles. The process begins with calibration, which involves analyzing the internal parameters of the camera. Subsequently, resection allows to determine the external parameters of the camera, and intersection calculates the ray coming from the 3D coordinate of the point in the projection. Searching for intersections or overlaps in photographs is necessary to obtain information about the 3D coordinates of a point. This process is called "collinearity" and is based on "collinearity equations," **(shown in Formula 1)** which demonstrates the geometric relationship where the center of the camera lens, the 3D coordinate location, and the corresponding 2D point on the camera's image sensor all lie on the same straight line (Guidu & Frischer, 2020).

$$x - x_0 = -c \, \frac{\left(R_{11}(X - X_0) + R_{21}(Y - Y_0) + R_{31}(Z - Z_0)\right)}{\left(R_{13}(X - X_0) + R_{23}(Y - Y_0) + R_{33}(Z - Z_0)\right)} \qquad \text{(a)}$$

$$y - y_0 = -c \, \frac{\left(R_{12}(X - X_0) + R_{22}(Y - Y_0) + R_{32}(Z - Z_0)\right)}{\left(R_{13}(X - X_0) + R_{23}(Y - Y_0) + R_{33}(Z - Z_0)\right)} \qquad \text{(b)}$$

Formula 1 for collinearity equations. They are used to identify the location of a point of a 3D object (X, Y, Z) relative to the point of its 2D image (x, y). Equation **(a)** calculates the x-coordinate, and **(b)** computes the y-coordinate. The symbol (*c*) estimates the focal length of the camera, and the negative sign indicates the direction of the image plan. (X_0, Y_0, Z_0) is the position of the camera center, and individual R_{11}, R_{13}, R_{21}, . . . , R_{33} are the rotation matrix elements representing the camera orientation.

Ultimately, this data is used to create a 3D cloud and, later, a 3D model. Types of photogrammetry, depending on the methods of obtaining the initial photographs, are mainly divided into aerial and terrestrial. Aerial photogrammetry is often practiced in digitization of architecture and makes use of unmanned drones, satellites, balloons, and airplanes to snap more photographs of the building. Terrestrial photogrammetry is applied to ground-based cameras; one of the most popular types of terrestrial photogrammetry is currently digital close-range photogrammetry. Other variations can also differ depending on the photographing method: for example, spherical photogrammetry employs a complete multi-image panorama instead of a single-frame photo.

The main advantage of photogrammetry is the accessibility and portability of the technology because even mobile phones can serve in creating a 3D model. In contrast to other scanning technologies, photogrammetry obtains color and texture data, which is a powerful instrument for archeologists. Additionally, it offers quick collection of metric data of objects of different sizes and complexity (Qader & Shwan, 2024). However, this technology has two main drawbacks. First, it is highly dependent on the camera, the quality of photo, and the necessary skills to shoot the right images, so that the simulation will take place without noise, problems with reflection, diffusion, light absorption, and the absence of fine details from the original scanned object. Second, it is more time consuming than other methods. To solve this problem, new algorithms have been developed, such as "Scale-Invariant Feature Transform" (Lowe, 2004) and "Speeded Up Robust Feature" (Bay et al., 2006), to significantly increase processing speed.

Modern photogrammetry pipelines involve Shape-from-Motion (SfM) workflows, which are integrated into computer softwares. SfM is a computational calculation of unmanaged and previously unknown coordinates from a group of overlapping images. This process additionally makes use of baked textures that include more details with less polycount, optimizing the photogrammetry output. In a recent case study, scientists Zachos and Anagnostopoulos (2024) used photogrammetry to digitize "Panagia Ekatontapiliani

Church." In another example, Dhanda et al. (2019) created VR environments using SfM photogrammetry; they tested this technique in digitization practices of Myin-pya-gu temple in Myanmar.

Structured Light

Structured light, another widely used active 3D scanning technique, was examined by Bell et al., (2016). It projects various light patterns, usually white or blue light, onto an object instead of a laser pulse. In general, projector and camera are needed for this scanning process. When the projector displays the patterns, the camera records their behavior and distance from the object's point to the camera lens. These patterns can include various data depending on the surface and geometry of the object, which allow a computer to collect the points to later visualize the shape.

This method is commonly used for digitizing medium to small size antiques with a high level of detail and geometric complexity. The main advantages of this technique are accuracy, high-resolution, and speed (Qader & Shwan, 2024). Depending on the equipment used, structured light may be incapable of capturing color (Ibid). However, in reality, this limitation could be addressed by the use of modern light scanners like Artec that already come with an RGB camera. Furthermore, Melendreras Ruiz et al. (2023) found that structured light generated a more detailed correlation with real colors compared to photogrammetry while scanning the "Holy Christ of the Blood."

Laser Triangulation

Laser triangulation is an active 3D scanning technique, meaning that it emits laser rays instead of relying on ambient light. This method was explained in great detail by Guidu and Frischer (2020). It implements a specialized laser scanner device. The scanner directs laser rays toward the surface of a particular object. When the rays reflect back, the scanner calculates their angle and return time, initially saving data. As the rays reflect, the angle of emission and reception constantly modifies, thus forming a triangle between the laser vertexes, the surface point, and the detector. Ultimately, the scanned data, or collected triangles, form polygons that literally build the geometry of the object: starting with a cloud mesh and ending with baked high-resolution textures, ready for export as a digital twin.

One of the main advantages of laser technology is its high precision and resolution, in addition to its ability to capture scanning data rapidly, in contrast to photogrammetry that requires a long processing time to preview results. The first laser triangulation device was named Cyrax 2400, capable of measuring 800 points per second with unclarity of 6 mm at a distance of 50 meters. Most laser devices now can record hundreds of thousands of data points in seconds, making them attractive tools for documentation of small objects.

Nevertheless, there are several disadvantages of laser triangulation. First, large or distant objects from a laser device are impractical to scan or capture due to sensor limits. Second, emitted laser rays are too sensitive to surface material: reflective, transparent, or dark surfaces can cause distortions, guaranteeing inaccurate results. Finally, the high cost of laser scanning equipment limits its adoption in non-technologically developed countries. Despite these limitations, the laser triangulation was used in the case study of scanning Mona Lisa, one of the most renowned artworks worldwide (Guidu & Frischer, 2020).

Computed Tomography

Computer Tomography (CT) is often used in cultural heritage to visualize a certain object's entire volume. This technique, as explained by Qader and Shwan (2024), presents an active 3D scanning method which does not use both laser or light, but x-ray source. CT was first built for usage in the medicine field. As of now, the same apparatus are also used in cultural heritage and are found especially helpful in reconstruction works.

By mirroring the x-rays from dissimilar angles and keeping the generated data, the CT can analyze and render hidden layers within the object. This method, in contrast to photogrammetry, laser, and structured light, offers an innovative feature to display internal details of the object. However, if the object is larger or smaller than the human body, this technique is unable to precisely capture it. Therefore, CT is mostly used for capturing humanlike sculptures.

Nonetheless, there is no CT-specialized equipment for non-human-like heritage. Moreover, the CT process presents some disadvantages such as micro defects, inaccessibility of scanning heritage that is not similar to the human body, and incompetence in scanning cultural heritage antiques made from metallic or solid materials like marble and stone. Japanese Kongo Rikishi statues were CT scanned, opening the door to analyzing their inner composition, particularly helpful for reconstruction efforts (Qader & Shwan, 2024).

LiDAR

LiDAR technology signifies Light Detection and Ranging. This technique uses light pulses to measure distance and generate point clouds of a particular object. As Tribhuvan (2024) described, this modern technology is a revolutionary achievement in the improvement of depth sensing. LiDAR is usually obtained with a laser, scanner, and GPS receiver and can be found in iPhone devices. In contrast to radar, which resembles the LiDAR, the latter makes use of light instead of radio waves. The laser emitter produces infrared short pulses of laser light and counts bounces back to the sensor. These pulses are a series of highly collimated, directional, coherent, and in-phase electromagnetic radiation that the system calculates based on flight time and captures the reflectivity of the surface (Zachos & Anagnostopoulos, 2024).

There are two methods of range determination in LiDAR: phase and Time-of-Flight (ToF). Also, Terrestrial LiDAR Scanning (TSL) is another form of digitization by using a tripod-mounted laser scanner (Benchekroun & Ullah 2021). The main advantages of LiDAR are accessibility and portability, relatively fast processing speed for archeological sites, and high quality of scanned models. However, there are several disadvantages: LiDAR itself without an RGB camera cannot detect the color. As a result of many point clouds, the device needs reliable storage and optimization systems. Oftentimes, multiple scanning positions are required to get comprehensive results, which is time-consuming. The LiDAR system was used in a forest in Rastatt, Germany and revealed a lot more data beneath the vegetation than other techniques. In another cultural heritage example, LiDAR was enacted to scan uncovered landscapes at Angkor Wat, Cambodia (Evans et al., 2013). A combination of LiDAR and close-range photogrammetry has been used to capture both the interior and exterior of architectural structures of the Great Wall of Ming Dynasty (Li et al., 2023).

Comparative Evaluation Table

Method	Applicable to	Advantages	Limitations
Photogrammetry	Small artifacts with high color fidelity	High-resolution texture alignment with cost-effective approach	Limited when scanning reflective surfaces and camera-dependent
Structured Light	Small to medium artifacts with highly complex geometry	High surface accuracy and more texture detail	Lacks color information if not connected to RGB camera
Laser Triangulation	Small artifacts with intricate and complex surface morphology	Quick scanning process and highly detailed scan	Device sensor is not suitable to scan large-scale objects
Computer Tomography	Human-like artifacts with internal layered structures	Clear and accurate internal visualization	Extensive device not suitable for metal or non-human-shaped forms
LiDAR	Large archeological environments including artifacts	High-precision scan results using a mobile device	Environmental particles cause model inaccuracies

Table 1. Comparative Evaluation of 3D Scanning Methods. Created by the author.

Applications

3D scanned heritage, or digital twins, enable multimodal documentation and presentation, including in both digital and physical formats. Multimodal documentation involves the use of multiple types of data capture, computational processing, and presentation techniques. This method not only focuses on 3D scans of antiques for documentation purposes, but it also extends to virtual museums that exhibit both tangible and intangible cultural heritage. Moreover, multimodal pipelines underscore the availability of resources to create both digital and physical replications of heritage, preserving the original antiques with a higher percentage of success than using solely digital or physical methods. While multimodal applications may differ in the techniques used, they should comprehensively justify methodological choices and ensure optimal digitization results. For example, as illustrated in **Figure 4**, an original artifact is 3D scanned using photogrammetry, and the resulting digital twin is applied in three interconnected applications for cultural preservation and presentation.

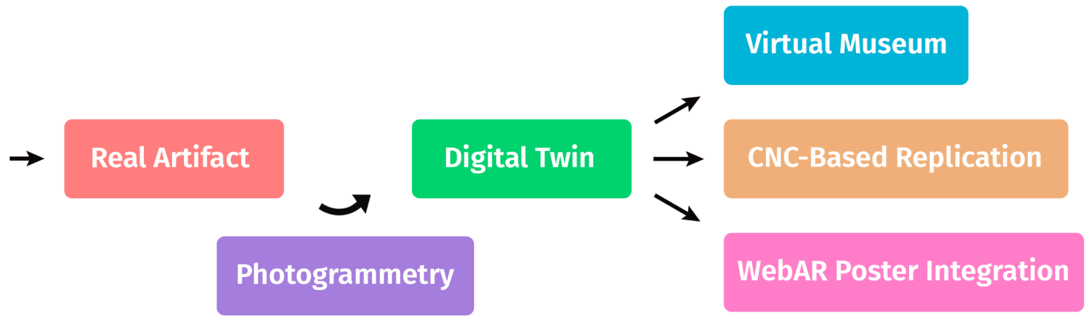

Figure 4. Multimodal Pipeline: From Artifact to Digital and Fabrication Products.
A diagram illustrating the multimodal digitization with both physical and digital use in various fields.
Created by the author.

Virtual Reality

Virtual Reality (VR) is a state-of-the-art technology that advances the living experience in computer worlds for users: various gadgets such as headsets, lenses, and screens render lifelike images in real time, allowing them to look around, transfer a space, and interact with objects similarly as they would in real life. In other words, this visually rich experience can be described as a life-like simulation that duplicates visible, auditory, and haptic sensory features of the real world, except for scent. VR consists of two important sides: the 3D world and connected level of interaction with realism (Paszkowska et al., 2022).

Initially designed by Ivan Sutherland in the 1960s, modern VR technologies support motion tracking, hand-tracking sensors, and haptic feedback systems, making the experience of diving into virtual reality even more lifelike. Additionally, computers were not powerful enough to render computer-generated environments at high speed and required significant computational power in the last century. In contrast, many modern VR headsets, Meta Quest 3 being one example, are capable of ultra-fast real-time rendering and can function as a portable device without a constant connection to a computer. While VR technologies are common in entertainment, education, simulation-training, healthcare, architecture, and marketing industries, their adoption in virtual museum environments is also progressively growing (TeamViewer, n.d).

VR museums embrace the concept of traditional museums by presenting collections of paintings, sculptures, and other types of artwork in digital space. These museum exhibitions are generally created from digital twins and collected in one virtual environment. The main feature of VR museums is next-level interaction with museum exhibitions, similar to real-life experience: visitors are able to virtually hold a specific antique in their hand; transform, rotate, and take a closer look at details of the object; and bring together virtual artifacts from a particular international collection into one scene. Also, using VR devices such as headset, glasses, and handheld controllers, they are able to not only view, explore, and perceive the artifacts but also to touch, move, and rotate them. In turn, this supports a better memorization of new information for museum visitors while displaying a more realistic world. One example of a VR museum is the National Gallery (Sainsbury Wing), which is available via the web on widely used VR devices such as the Oculus Meta Quest.

Augmented Reality

Augmented Reality (AR) is an advanced and distinctive technology that enhances the real-world setting by adding and blending digital visual information on top of the existing world. AR does not completely recreate the world as it does VR, but it still preserves the realistic image as a lower layer. AR has been used in the development of information and communication technologies (ICTs) since the 1990s.

The main fields of AR implementation include architecture, education, medicine, automation, and also cultural heritage. The primary functions of AR in the cultural heritage field are: enhancing viewer engagement; improving education and learning experiences; aiding preservation and documentation; bringing historical architecture back to life; and supporting digital technologies. AR applications are usually developed for both mobile and web platforms by designing user interface (UI) elements, modeling 3D representational models, and programming surface recognition functions. Game engines such as Unity and Unreal Engine support AR development by default.

Although VR technology is more technically complex and costly for ordinary users, AR provides a more creative and accessible means of presenting cultural heritage. For example, during the Galloway Hoard exhibition, the visitors of National Museum of Scotland scanned QR posters to view various digital twins of Viking artifacts on their phone (Coulson, 2022). In another example, a mobile AR application developed by researchers presented a 3D reconstruction of the “Prejmer Fortified Church” in Romania (Boboc et al., 2022).

Computer Numerical Control

Computer Numerical Control (CNC) is a complex system of controlling fabrication machines using computers. The work of such machines was traditionally carried out manually by craftsmen, but in the modern era, the cutting-edge automation machines are available to increase the speed and quality of the process while reducing its physically demanding labor. Using CNC machines is becoming common, but it is still a technically complicated process that needs professionals to manufacture.

In the field of cultural heritage, Islami and Sutanto outlined a process of making wood carving in two methods: manual and digital using CNC. While manual production is understandable and includes traditional carving instruments, digital production represents an innovative approach: utilizing computer software and Joint Photographic Experts Group (JPG) media files, Islami and Sutanto generated a Drawing Exchange Format (DXF) file. This DXF file was subsequently converted into a Computer-Aided Design (CAD) model to write machine-readable code, commonly known as G-code. Later, simulation tests were programmed using computer software to preview the ultimate results before running the program on a milling machine.

The most important stages of the CNC-based manufacturing process that directly determine the precision of the final carved model include: finding the initial point and dimensions of the model, programming a secure and planned toolpath, creation of identical cutting tools, and the correction of machining parameters. By carefully following these aspects and using a high-quality digital twin, computer specialists can produce accurate physical replicas of cultural heritage antiques.

Conclusion

This study presented the implementation of 3D scanning to preserve endangered cultural heritage and highlighted the advantages of digitization over traditional methods. The ideology of the digital twin was established to describe the result of 3D scanning. An extensive description of each scanning technique, such as photogrammetry, structured light, or computer tomography, was introduced. These descriptions were accompanied by practical case studies from researchers, such as the case study of scanning the Mona Lisa. Moreover, visual figures illustrated the contrast between different 3D scanning methods and evaluated their applicability for optimal use. In addition, modern applications of digital twins, such as VR museums, were explored as means to preserve, educate, and promote cultural heritage. In conclusion, the multimodal pipeline, containing both digital and physical fabrication of digital twins using CNC machines, was expounded. Future work will include a practical case study of the digitization of a cultural object using 3D scanning and modeling technologies.

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